24BCA52	Backend Development	45 Hours

Course Objectives:

CO1: Understand the fundamental concepts of backend development, client-server architecture, HTTP protocols, and REST API design.

CO2: Develop knowledge of building server-side applications using Python (Flask) and handling requests, responses, and JSON data exchange.

CO3: Understand database integration and perform CRUD operations through backend APIs with proper data handling and validation.

CO4: Gain knowledge of backend development using Node.js (Express) and asynchronous programming for scalable application development.

CO5: Understand modern backend tools including no-code/low-code platforms (e.g., n8n), AI-assisted (vibe coding) development, and integration of AI applications such as chatbots.

CO6: Learn deployment, version control, and production practices along with building AI-powered workflows and agent-based backend systems.

Prerequisite:

Basic knowledge of programming concepts including variables, loops, functions, and data structures. Familiarity with fundamental web concepts such as client-server architecture, HTTP/HTTPS, and APIs. Basic understanding of JavaScript and any programming language (Python preferred). Awareness of databases (DBMS concepts) and willingness to learn modern development tools, automation platforms, and AI-assisted coding techniques.

Course Content:

UNIT I: Backend Fundamentals and REST API Basics: **[12 Hours]**

Server-Side Concepts and API Foundations: Introduction to backend and client-server architecture. HTTP methods, request-response cycle. Designing simple REST APIs and understanding endpoints. Building APIs using Python (Flask): Creating basic REST APIs using Flask. Routing, handling requests/responses, returning JSON data.

UNIT II: Database Integration and Backend Development: **[11 Hours]**

Database and CRUD Operations: Introduction to databases and integration. Performing CRUD operations using backend APIs. Handling user input and basic validation. Backend using Node.js and Express: Introduction to Node.js and Express. Creating REST APIs, middleware usage, and handling asynchronous operations.

UNIT III: Modern Backend Tools and AI-Assisted Development: **[11 Hours]**

No-Code/Low-Code Backend and Automation Tools: Using n8n for building backend workflows, API integrations, and automation without heavy coding. AI-Assisted Development (Vibe Coding): Using AI tools to generate backend logic, APIs, and integrations. Writing minimal prompts, refining outputs, and debugging AI-generated code.

UNIT IV: AI Applications, Agents and Deployment: **[11 Hours]**

Building AI Applications and Chatbots: Creating simple AI-based applications and chatbots using APIs. Integrating backend with AI services for automation and user interaction. Agents, Deployment and Production Practices: Introduction to AI agents and workflow-based systems. Deployment of backend services using cloud platforms. Basic version control and production practices.

Backend Textbooks:

1. Hillar, Gastón C., Building RESTful Python Web Services, Packt Publishing, 2016.
2. Brown, Ethan, Web Development with Node and Express: Leveraging the JavaScript Stack, O'Reilly Media, 2014.
3. Kumar, P. Sivakumar and others, AI Agents Using n8n for Beginners, CSharpCorner Publishing, 2025.
4. Subramanian, Harihara and Raj, Pethuru, RESTful API Design Patterns and Best Practices, O'Reilly Media, 2025.

Resources

Backend Fundamentals & REST APIs

1. https://developer.mozilla.org/en-US/docs/Web/HTTP
2. https://restfulapi.net/

Python Backend (Flask)

1. https://flask.palletsprojects.com/en/stable/quickstart/
2. https://docs.python.org/3/tutorial/
3. https://realpython.com/tutorials/flask/

Node.js and Express.js

1. https://nodejs.org/en/docs
2. https://expressjs.com/en/starter/installing.html
3. https://www.freecodecamp.org/news/tag/nodejs/

Database and CRUD Operations

1. https://developer.mozilla.org/en-US/docs/Learn/Server-side/Express_Nodejs/mongoose
2. https://www.w3schools.com/sql/
3. https://www.mongodb.com/docs/

No-Code / Automation Tools

1. https://docs.n8n.io/
2. https://zapier.com/learn/
3. https://make.com/en/help

AI Applications and Chatbots

1. https://platform.openai.com/docs
2. https://python.langchain.com/docs/
3. https://learn.microsoft.com/en-us/azure/ai-services/

Version Control and Deployment

1. https://git-scm.com/docs
2. https://docs.github.com/en/get-started